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Continuous recliner of vehicle seats

Abstract

Provided is a continuous recliner for vehicle seats, in which an internal gear and a cover plate are fixed to each other, and the cover plate is welded to a cushion bracket. A heat transfer path is lengthened to prevent deterioration in the surface roughness of sliding friction parts, thereby reducing the operating force and operating noise of the recliner.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Korea Patent Application No. 10-2022-0025728, filed Feb. 28, 2022, the entire contents of which is incorporated herein for all purposes by this reference.

FIELD

(2) The present disclosure relates to a continuous recliner for vehicle seats, and relates to a continuous recliner for vehicle seats capable of transferring a rotational force, performing deceleration and maintaining an angle-adjusted state using an external gear and an internal gear.

BACKGROUND

(3) A recliner is installed at a connection portion between a seatback and a seat cushion in a vehicle seat. The recliner is a device capable of enabling the angle of the seatback to be adjusted. In an unlocked state, the angle of the seatback is adjusted forward and backward, and in a locked state, the angle of the seatback, which has been adjusted to a desired position, is maintained.

(4) As shown in FIG. 1, a recliner **10** is installed between a back bracket **20** mounted at the lower end of a side of a seatback frame **40** and a cushion bracket **30** mounted at the rear end of a side of a seat cushion frame (not shown).

(5) FIG. 2 is a cross-sectional view of the recliner **10**. An external gear **11** and an internal gear **12** are eccentrically inscribed, and a bush **14a** is press-fitted into an inner diameter portion of the internal gear **12**.

(6) A socket **13** is rotatably inserted into an inner diameter portion of a center boss of the external gear **11**, and a pair of wedges **14b** and a cam sleeve **14c** are installed between the bush **14a** and an outer diameter portion of the center boss of the external gear **11**. The bush **14a**, the wedges **14b**, and the cam sleeve **14c** are generally referred to as sliding friction parts **14** because their contact surfaces slide to cause friction when the recliner is operated.

- (7) The boss of the internal gear **12** is inserted into and welded to the inner circumferential surface of an installation hole **31** of the cushion bracket **30**.
- (8) A disc-like cover plate **16** is welded to the inner diameter portion of the boss of the internal gear **12** to prevent the socket **13**, the sliding friction parts **14**, and a wedge spring **15** from escaping to the outside of the recliner **10**. A guide ring **17** installed on an outer periphery of the external gear **11** prevents separation between the external gear **11** and the internal gear **12**.
- (9) The wedges **14b** are pushed by the wedge spring **15** installed on the outside of the socket **13** to constraint rotation of the external gear **11** with respect to the internal gear **12** by a wedge action.
- (10) The rotation of the socket **13** causes the wedges **14b** to rotate in the opposite direction to a locked position via the cam sleeve **14c** to release a constrained state of the external gear **11** due to the wedge **14b** and eccentrically rotate the external gear **11** with respect to the internal gear **12** by being interlocked with the rotation of the socket **13**. Therefore, the rotation (self-rotation) of the external gear **11** according to a gear ratio between the internal gear **12** and the external gear **11** is caused, thereby adjusting the angle of the seatback connected to the side of the external gear **11**.
- (11) On the other hand, in the conventional recliner **10** as described above, the boss (central cylindrical portion) of the internal gear **12** is necessarily elongated to protrude by 15 mm or more in order to weld the internal gear **12** to the cushion bracket **30**. The boss is formed by burring processing. A process of forming the boss of 15 mm or more with burring processing is difficult and time-consuming. Therefore, many defects occur in which variability in protrusion length of the boss is large and the end shape is not perfect, and the production speed of the internal gear **12** is reduced.
- (12) In addition, since the internal gear **12** is welded to the cushion bracket **30** and the cover plate **16** is welded to the internal gear **12**, excessive welding heat is transferred to the sliding friction parts **14** via the internal gear **12**. Therefore, the sliding friction parts **14** are damaged by heat and the surface roughness is deteriorated to cause increased friction loss during operation of the recliner **10**, resulting in an increase in operating force and operating noise of the recliner.
- (13) The use of laser welding to reduce the thermal effect as described above causes an increase in manufacturing cost, and nevertheless, welding defects often occur during laser welding due to variability in length of the boss of the internal gear **12** and defective end shape as described above.

SUMMARY

- (14) Therefore, the present disclosure has been made to solve the above problems, and an object of the present disclosure is to provide a continuous recliner for vehicle seats, in which thermal damage to sliding friction parts is reduced to reduce an operating force and an operating noise, there is no need to form a long boss for an internal gear to improve the production speed of the internal gear as well as reduce defects, a wide choice of welding types is provided and welding defects are reduced.
- (15) According to an aspect, a continuous recliner for vehicle seats includes an external gear fixed to a back bracket of a seatback frame, an internal gear eccentrically inscribed in the external gear, and a cover plate press-fitted into and coupled to the internal gear and fixed to a cushion bracket of a seat cushion frame.
- (16) The cover plate may include a disc portion forming one side surface of the recliner, a cylindrical portion extending from an end of the disc portion in a direction of a central axis of the recliner, and a flange portion outwardly extending from an end of the cylindrical portion in a radial direction of the recliner.
- (17) The cylindrical portion of the cover plate may be inserted into and welded to an installation hole formed in the cushion bracket.
- (18) The flange portion of the cover plate may be formed larger than an outer diameter of the internal gear, the flange portion including a rim portion having an extended thickness at an end of the flange portion, and the internal gear may be press-fitted into an inner diameter surface of the rim portion.
- (19) The inner diameter surface of the rim portion of the cover plate may be formed with gear teeth

to mesh with gear teeth formed in an outer diameter portion of the internal gear.

(20) The flange portion of the cover plate may be formed with gear teeth at an end thereof to mesh with gear teeth formed in an inner diameter portion of the internal gear.

(21) A guide ring may be mounted on the external gear, and an end of the guide ring on an inner diameter side thereof may be formed to have a length extending beyond a boundary between the internal gear and the cover plate to cover a side of the flange portion of the cover plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view showing a general recliner installation structure.

(2) FIG. 2 is a cross-sectional view showing a coupling structure between a recliner and a cushion bracket according to the prior art.

(3) FIG. 3 is an exploded perspective view of a recliner according to a first embodiment of the present disclosure.

(4) FIG. 4 is a rear view of an internal gear and a cover plate in FIG. 3 and a view showing a coupling state between the internal gear and the cover plate.

(5) FIG. 5 is a cross-sectional view showing a coupling structure between the recliner and the cushion bracket according to the first embodiment.

(6) FIG. 6 is an exploded perspective view of a recliner according to a second embodiment of the present disclosure.

(7) FIG. 7 is a cross-sectional view showing a coupling structure between the recliner and the cushion bracket according to the second embodiment.

DETAILED DESCRIPTION

(8) The present disclosure may be variously modified and may have various embodiments, and particular embodiments illustrated in the drawings will be described in detail below. However, the description of the exemplary embodiments is not intended to limit the present disclosure to the particular exemplary embodiments, but it should be understood that the present disclosure is to cover all modifications, equivalents and alternatives falling within the spirit and technical scope of the present disclosure. Thicknesses of lines illustrated in the accompanying drawings, sizes of constituent elements, or the like may be exaggerated for clarity and convenience of description.

(9) In addition, the terms used below are defined in consideration of the functions in the present disclosure and may vary depending on the intention of a user or an operator or precedents.

Therefore, the definition of the terms should be made based on the entire contents of the present specification.

(10) Hereinafter, preferred embodiments according to the present disclosure will be described in detail with reference to the accompanying drawings. Since the basic configuration of a continuous recliner for vehicle seats according to the present disclosure is the same as that of the prior art, a description will be given while the same reference numerals are denoting components. Also, in the drawings, a triangle indicates a welded portion.

(11) That is, a recliner **10** according to the present disclosure may include an external gear **11** and an internal gear **12** which are eccentrically inscribed, sliding friction parts **14** (a bush **14a**, a wedge **14b**, and a cam sleeve **14c**) installed between the external gear **11** and the internal gear **12** to engage or release the external gear **11** to or from the internal gear **12**, and a socket **13** rotatably installed on a center boss of the external gear **11** to operate the wedge **14b**.

(12) In addition, the recliner **10** according to the present disclosure may further include a wedge spring **15** installed on an outer portion of the socket **13** to continuously apply a force in a direction constraining the wedge **14b**, a cover plate **16** configured to block the central portion of one side of the recliner to prevent escapes of internal parts (the socket **13**, the sliding friction parts **14**, and the

wedge spring **15**), and the guide ring **17** configured to prevent the separation between the external gear **11** and the internal gear **12**.

(13) The recliner **10** according to the present disclosure is characterized by its coupling structure with the cushion bracket **30**. That is, the recliner **10** according to the present disclosure is characterized in that the recliner **10** is mounted to the cushion bracket **30** in such a way that the cover plate **16** is welded to the cushion bracket **30**.

(14) In addition, as the cover plate **16** is coupled to the internal gear **12** by press fitting, the internal gear **12** is mounted on the side of the cushion bracket **30** via the cover plate **16** to cause the internal gear **12** to be fixed in a non-rotatable state.

(15) Two embodiments including all of the above features will be described in detail below.

(16) FIGS. **3** to **5** are views showing a first embodiment.

(17) A cover plate **16** according to the first embodiment may include a disc portion **16-1** disposed on the outer side of a socket **13** in an assembled state of a recliner **10**, a cylindrical portion **16-2** bent and extended from an end of the disc portion **16-1** in a direction parallel to the direction of the central axis of the recliner **10**, a flange portion **16-3** outwardly extended from an end of the cylindrical portion **16-2** in the radial direction of the recliner **10**, and a rim portion **16-4** having an extended thickness at an end of the flange portion **16-3**.

(18) The flange portion **16-3** may be formed larger in the radial direction than the internal gear **12** in the assembled state of the recliner, and the rim portion **16-4** may be formed around the entire periphery of the plate **16**, that is, the entire periphery of the cover plate **16**.

(19) The rim portion **16-4** may be formed to be thicker than the flange portion **16-3**, and the internal gear **12** is press-fitted to the inner circumferential surface of the rim portion **16-4**, so that the internal gear **12** and the cover plate **16** are fixed to each other. That is, the internal gear **12** and the cover plate **16** are fixed by press-fitting so as for the internal gear **12** and the cover plate **16** not to move in the axial and circumferential directions. In this case, in order to more reliably prevent rotational movement between the internal gear **12** and the cover plate **16**, gear teeth **16a** corresponding to gear teeth **12a** formed in the outer circumferential surface of the internal gear **12** may be formed in the inner circumferential surface of the rim portion **16-4**. In this case, the gear teeth **12a** of the internal gear **12** and the gear teeth **16a** in the cover plate **16** are engaged with each other to certainly prevent relative movement (rotation) between the internal gear **12** and the cover plate **16** in the circumferential direction. That is, the internal gear **12** is fixed to the cover plate **16** to be unable to rotate.

(20) The cover plate **16** may be mounted to the cushion bracket **30** in such a way that one side of the cylindrical portion **16-2** is welded to the inner circumferential surface of the installation hole **31** of the cushion bracket **30**. Since the cushion bracket **30** is mounted to the cushion frame and fixed in position, both the cover plate **16** welded to the cushion bracket **30** and the internal gear **12** press-fitted into the cover plate **16** are fixed, and the external gear **11** rotates eccentrically around the external circumference of the internal gear **12**.

(21) When the internal gear **12** is approximately divided to two portions with respect to the thickness thereof, the gear teeth **16a** of the cover plate **16** mesh with one half portion of the internal gear **12**, and the gear teeth **11a** of the external gear **11** mesh with the other half portion of the internal gear **12**.

(22) In the assembled state of the recliner, the disc portion **16-1** of the cover plate **16** may block one side of the recliner **10** to prevent the inner socket **13**, the sliding friction parts **14**, the wedge spring **15** or the like from escaping to the outside.

(23) The cylindrical portion **16-2** may provide a welding surface to which the cushion bracket **30** is able to be welded, and form a stepped portion between the disc portion **16-1** and the flange portion **16-3**.

(24) The flange portion **16-3** may expand the radius of the cover plate **16** more than the radius of the internal gear **12**, and provide a support surface when the internal gear **12** is press-fitted into the

rim portion **16-4**.

(25) The rim portion **16-4** may provide a press-fitting surface into which the outer diameter portion of the internal gear **12** is press-fitted, and may be formed with the gear teeth **16a** which mesh with the gear teeth **12a** of the internal gear **12** to reliably suppress rotation of the internal gear **12** with respect to the cover plate **16**. In addition, the guide ring **17** welded to an end of the outer diameter portion of the external gear **11** is installed in a structure surrounding the rim portion **16-4** to prevent the external gear **11** and the internal gear **12** from being separated from each other.

(26) In the recliner according to the first embodiment, the cover plate **16** fixedly coupled to the internal gear **12** is welded to the cushion bracket **30** instead of the internal gear **12**, thus removing a necessity to perform a burring processing for generating a long boss on the internal gear **12**. Therefore, the occurrence of processing defects such as variability in length of the boss or defective end shape is fundamentally prevented, and the manufacturing of the internal gear **12** is easy to improve production speed.

(27) In addition, since welding heat occurring at a welding portion between the cover plate **16** and the cushion bracket **30** is transferred to the sliding friction parts **14** via the cover plate **16** and the internal gear **12**, a heat transfer path is increased compared to the prior art, and thus the amount of heat transferred to the sliding friction parts **14** is reduced. Therefore, since the deterioration in surface roughness due to thermal damage to the sliding friction parts **14** does not occur, friction is reduced when the recliner is operated, thus reducing not only the operating force of the recliner but also the operating noise of the recliner.

(28) As described above, since the safety of internal parts against welding heat is improved, not only laser welding but also other types of welding, such as MAG (Metal Active Gas) welding, may be performed. Therefore, welding cost is reduced and laser welding that requires high precision for an object to be welded is not required, thereby preventing welding defects due to variability in length of the boss of the internal gear **12** and the defective shape.

(29) FIGS. **6** and **7** show a second embodiment of the present disclosure. In the second embodiment, the present disclosure is applied when the external gear **11** and the internal gear **12** are coupled in a double-stepped structure.

(30) In order to improve the coupling stability of the external gear **11** and the internal gear **12** and reduce the load applied to the gear teeth, the external gear **11** and the internal gear **12** of the recliner may have a multi-stepped coupling structure, and a double-stepped coupling structure is usually applied as shown in FIGS. **6** and **7**.

(31) For the double-stepped coupling structure, two rows of gear teeth **11a** and **11a'** are continuously formed in the inner circumferential surface of the external gear **11** while a stepped portion is provided therebetween. Correspondingly, two rows of gear teeth **12a** and **12a'** are continuously formed in the outer circumferential surface of the internal gear **12** while a stepped portion is provided therebetween. Generally, since second-stepped gear teeth **12a'** (inner gear teeth in radial direction) of the internal gear **12** are formed by press processing, the inner circumferential surface of a stepped portion in which first-stepped gear teeth **12a** (outer gear teeth in radial direction) are formed, are formed with gear teeth **12b** having the same shape as the second-stepped gear teeth **12a'**.

(32) The gear teeth **16b** formed in the outer circumferential surface of the flange portion **16-3** of the cover plate **16** are press-fitted into the gear teeth **12b**, so that the internal gear **12** and the cover plate **16** are mutually fixed in the axial and circumferential directions.

(33) The disc portion **16-1** and the cylindrical portion **16-2** of the cover plate **16** according to the second embodiment are the same as those of the first embodiment, and the flange portion **16-3** is formed shorter than that of the first embodiment in accordance with the formation position of the gear teeth **12b** on an inner diameter side of the internal gear **12**. The gear teeth **16b** press-fitted between the teeth **12b** of the internal gear **12** are formed at an end of the flange portion **16-3**.

(34) Even in the case of this second embodiment, the internal gear **12** and the cover plate **16** are

fixed to each other in the axial and circumferential directions by press-fitting and tooth engagement, and the cylindrical portion **16-2** of the cover plate **16**, not the internal gear **12**, is inserted into and welded to the installation hole **31** of the cushion bracket **30**, thus achieving the same effect as that of the first embodiment.

(35) That is, since a boss is not formed in the internal gear **12**, reduction of defects and improvement of production speed are achieved during the production of the internal gear **12**, and damage to internal parts due to welding heat is prevented, thereby resulting in same effects as the first embodiment, such as reduction in the operating force and operating noise of the recliner. Since the mechanism (operating principle or structure) for effect generation is the same as that of the first embodiment, redundant description will be omitted.

(36) On the other hand, in the second embodiment, an inner end of the guide ring **17** in the radial direction is molded to extend to the side of the flange portion **16-3** of the cover plate **16** beyond the boundary between the internal gear **12** and the cover plate **16**. Therefore, the guide ring **17** is caught on the flange portion **16-3**, thus preventing the internal gear **12** and the cover plate **16** from being separated from the external gear **11** in the axial direction.

(37) According to the present disclosure as described above, it is possible to provide a continuous recliner having a structure in which the cover plate coupled to the internal gear is welded to the cushion bracket.

(38) Therefore, it is possible to remove a necessity to form a boss for welding to the cushion bracket in the internal gear and facilitate processing of the center hole of the internal gear to improve the production speed of the internal gear and reduce shape defects.

(39) In addition, since welding heat is first transferred to the cover plate and then to the internal gear and transferred to the sliding friction parts from the internal gear, the amount of welding heat transferred to the sliding friction parts is reduced. Therefore, the degradation in surface roughness due to thermal damage to the sliding friction parts does not occur, and thus, the frictional force is reduced during operation of the sliding friction parts, thereby reducing the operating force and operating noise of the recliner.

(40) Since the safety of the sliding friction parts against the welding heat is improved as described above, there is no need to use laser welding alone when the recliner is welded to the cushion bracket. That is, not only laser welding but also other types of welding may be performed.

(41) In addition, since problems such as variability in the length of the boss or shape defects do not occur in the internal gear, frequent welding defects does not occur.

(42) Although some embodiments have been provided to illustrate the invention in conjunction with the drawings, it will be apparent to those skilled in the art that the embodiments are given by way of illustration only, and that various modifications and equivalent embodiments can be made without departing from the spirit and scope of the invention. Accordingly, the scope and spirit of the present disclosure should be limited only by the following claims.

(43) TABLE-US-00001 REFERENCE NUMERALS 10: Recliner 11: External gear 11a and 11a': Gear teeth 12: Internal gear 12a, 12a', 12b: Gear teeth 13: Socket 14: Sliding friction part 14a: Bush 14b: Wedge 14c: Cam sleeve 15: Wedge spring 16: Cover plate 16a and 16b: Gear teeth 16-1: Disc portion 16-2: Cylindrical portion 16-3: Flange portion 16-4: Rim portion 17: Guide ring 20: Back bracket 30: Cushion bracket 31: Installation hole 40: Seatback frame

Claims

1. A continuous recliner for vehicle seats comprising: an external gear fixed to a back bracket of a seatback frame; an internal gear eccentrically inscribed in the external gear; and a cover plate press-fitted into and coupled to the internal gear and fixed to a cushion bracket of a seat cushion frame, the cover plate comprising: a disc portion forming one side surface of the recliner; a cylindrical portion extending from an end of the disc portion in a direction of a central axis of the

recliner, the cylindrical portion of the cover plate being inserted into and welded to an installation hole formed in the cushion bracket; and a flange portion outwardly extending from an end of the cylindrical portion in a radial direction of the recliner.

2. The continuous recliner of claim 1, wherein the flange portion of the cover plate is formed with gear teeth at an end thereof to mesh with gear teeth formed in an inner diameter portion of the internal gear.

3. The continuous recliner of claim 2, wherein a guide ring is mounted on the external gear, and an end of the guide ring on an inner diameter side thereof is formed to have a length extending beyond a boundary between the internal gear and the cover plate to cover a side of the flange portion of the cover plate.

4. The continuous recliner of claim 1, wherein the flange portion of the cover plate is formed larger than an outer diameter of the internal gear, the flange portion comprising a rim portion having an extended thickness at an end of the flange portion, and wherein the internal gear is press-fitted into an inner diameter surface of the rim portion.

5. The continuous recliner of claim 4, wherein the inner diameter surface of the rim portion of the cover plate is formed with gear teeth to mesh with gear teeth formed in an outer diameter portion of the internal gear.
